

Lecture III - Building Reusable Functions

Programming with Python

Dr. Tobias Vlček

Kühne Logistics University Hamburg - Fall 2026

Checkpoint 1

The quarterly board review. The first 40 minutes are the checkpoint. It starts now.

- **Individual work:** no AI, no neighbors, no chat
- The **link and QR** are handed out in class. Open it and start
- ~6 short tasks: write code, trace code, fix a bug, answer a multiple choice
- It sweeps **Sessions I-II**: variables, types, `round` and f-strings, `if/elif`, `for` loops

When you're done

Menu → *Download* → *Download Python code* → upload the `.py` to the “Checkpoint 1” assignment on Moodle, **before the 40 minutes are up**. No retakes: one sitting.

...

Note

The green live checks are **provisional**. The final grading runs on my side. And take a breath: everything in it was rehearsed in the labs.

Episode 3: The Copy-Paste Soup

After the board review

Pens down. The checkpoint is behind you. Now the reason we're all here.

...

Tobi has been “reusing” code the only way he knows how: he pasted the same receipt block **14 times**, once per menu item. A ten-cent price change yesterday cost him a whole afternoon of hunting down copies.

...

Today the code learns to **reuse itself**. We meet the function, and build a first class.

Writing Functions

The copy-paste pain

Tobi totals each receipt line by hand, the same shape, over and over:

```
print(round(2 * 4.50, 2))    # 2 wraps
print(round(1 * 12.00, 2))  # 1 bowl
print(round(5 * 3.20, 2))   # 5 fries
```

```
9.0
12.0
16.0
```

...

Same calculation, retyped every time. Change the rule once and you must fix it everywhere. There is a better way.

def: name the calculation once

Write the calculation **once**, give it a name, and call it whenever you need it:

```
def line_total(qty, price):    # def NAME(inputs):
    return round(qty * price, 2) # indented body, hands a value back

print(line_total(2, 4.50))
```

```
9.0
```

...

`def` starts the definition · the name is `line_total` · the block is **indented** · `return` hands the answer back.

Parameters vs arguments

- **Parameters** are the names in the definition, the function's inputs: `qty`, `price`
- **Arguments** are the actual values you pass when you **call** it: `2`, `4.50`

```
def line_total(qty, price):    # qty and price are PARAMETERS
    return round(qty * price, 2)

print(line_total(3, 3.20))     # 3 and 3.20 are ARGUMENTS
```

```
9.6
```

...

Same function, different arguments, different result. No retyping.

Predict: which number is the bill?

Tobi wants a **40 % tip on a 10 EUR bill**... no wait, a **10 % tip on 40 EUR**. He types it in a hurry. What prints?

```
def with_tip(bill, percent):
    return round(bill * (1 + percent / 100), 2)

print(with_tip(10, 40))
```

a) 44.0 b) 14.0 c) Error

...

Predict first. Pick a letter, then I reveal the answer.

Answer: position decides, not intent

b) 14.0: arguments land on parameters **by position**. 10 becomes `bill`, 40 becomes `percent`: a 40 % tip on 10 EUR. Python cannot read what Tobi meant, only where he put it.

```
def with_tip(bill, percent):
    return round(bill * (1 + percent / 100), 2)

print(with_tip(10, 40))    # bill=10, percent=40
print(with_tip(40, 10))   # bill=40, percent=10
```

14.0
44.0

`return`: hand the value back

`return` sends a value **back to whoever called** the function, so you can store it and use it later:

```
def line_total(qty, price):
    return round(qty * price, 2)

subtotal = line_total(3, 3.20)    # catch what came back
print(subtotal)
print(subtotal + 1.50)           # ...and keep using it
```

9.6
11.1

Predict: print or return?

`label_price` **prints** but has no `return`. What does the **last line** show?

```
def label_price(price):
    print(f"{price:.2f} EUR")
```

```
result = label_price(8.50)
print(result)
```

a) 8.50 EUR b) Error c) None

...

Predict first. Pick a letter, then I reveal the answer.

Answer: `print` shows, `return` hands back

c) **None:** `label_price` prints its line while running, but with no `return` it hands back `None`. That `None` is what lands in `result`. Printing is not returning.

```
def label_price(price):
    print(f"{price:.2f} EUR")

result = label_price(8.50)  # prints while running...
print(result)              # ...but the value handed back is None
```

```
8.50 EUR
None
```

Your turn: 5-10 minutes

Open the exercise (scan the QR or type the link):

python.tobiasvlcek.com/notebooks/ex_03_a/



First **predict** what happens, then run it.

Default arguments

A parameter can carry a **default**, used when the caller leaves it out:

```
def service_fee(total, rate=0.05):    # rate defaults to 5 %
    return round(total * rate, 2)
```

```
print(service_fee(80))                # default rate → 4.0
print(service_fee(80, 0.10))          # override it → 8.0
```

```
4.0
8.0
```

...

Defaults let one function cover the common case and the special case.

Functions can call functions

A function can use another function you already wrote (reuse builds on reuse):

```
def bill(qty, price):
    items = line_total(qty, price)    # reuse line_total
    return round(items + service_fee(items), 2) # ...and service_fee

print(bill(2, 4.50))
```

```
9.45
```

...

`line_total` and `service_fee` do their jobs; `bill` just orchestrates. That's how small pieces become a program.

Your turn: 5-10 minutes

Open the exercise (scan the QR or type the link):

python.tobiasvlcek.com/notebooks/ex_03_b/



First **predict** what happens, then run it.

Scope, then a First Class

Local names stay local

A parameter is the function's **own private copy**. What happens inside stays inside:

```
def bump(n):  
    n = n + 10      # changes the function's OWN copy of n  
    return n  
  
print(bump(3))      # 13, the returned value
```

13

...

Names created inside a function live in its **scope**. They vanish when the function ends.

Predict: does the global move?

`bump` adds ten to its parameter. We call it, then print `stock`. What does the **last line** print?

```
def bump(n):  
    n = n + 10  
    return n  
  
stock = 3  
bump(stock)  
print(stock)
```

a) 3 b) 13 c) Error

...

Predict first. Pick a letter, then I reveal the answer.

Answer: the outside variable is untouched

a) `3`: `bump` changes only its own copy `n`. We never stored what it returned, so `stock` out here never moves. The function cannot reach out and rewrite your variables.

```
def bump(n):  
    n = n + 10  
    return n  
  
stock = 3  
bump(stock)      # the return value is thrown away  
print(stock)     # still 3
```

3

Why that's a feature, not a limit

- A function can't **secretly** change your variables, no spooky action at a distance
- You can call it a hundred times and trust your data stays put
- To use a result, you **catch the return** (`stock = bump(stock)`). Deliberate, visible

...

Isolation is what makes functions **safe to reuse**. That is the whole point of this episode.

A class bundles data with behavior

Sometimes data and the things you do with it belong **together**. A **class** is a blueprint that bundles both:

- **data**: the facts about one thing (a courier's name, their distance)
- **behavior**: what you can compute from it (their fee)

...

A **method** is just a function that lives inside a class and can read the object's own data.

class, __init__, and self

```
class Delivery:  
    def __init__(self, courier, distance_km): # runs when you build one  
        self.courier = courier                # store data ON the object  
        self.distance_km = distance_km  
        self.rate_per_km = 1.20              # every delivery carries its rate  
  
    def fee(self):                             # a method: note self  
        return round(self.distance_km * self.rate_per_km, 2)
```

...

`__init__` sets up a new object · `self` is *this* object · attributes are stored on `self` and read back through `self.fee()` multiplies **two** stored attributes.

Build one, call its method

Instantiate the class to make an object, then use its data and its methods:

```
trip = Delivery("Nadia", 4)    # build one, __init__ runs

print(trip.courier)            # read stored data
print(trip.fee())              # call the method
```

```
Nadia
4.8
```

...

`trip` is one `Delivery` object. `trip.fee()` computes from *its own* stored distance: data and behavior, traveling together.

Predict: does the second delivery overwrite the first?

Two deliveries get built from the same class, one after the other. What does the **last line** print?

```
trip = Delivery("Nadia", 4)
second = Delivery("Omar", 10)

print(trip.fee())
```

a) 12.0 b) Error c) 4.8

...

Predict first. Pick a letter, then I reveal the answer.

Answer: every object carries its own data

c) 4.8: `__init__` ran twice and stored the data on **two different objects**. `self` inside `trip.fee()` is `trip`, so it reads Nadia's 4 km, not Omar's 10. Instances do not share attributes.

```
trip = Delivery("Nadia", 4)
second = Delivery("Omar", 10)

print(trip.fee())    # self is trip: 4 km
print(second.fee())  # self is second: 10 km
```

```
4.8
12.0
```


One honest slide about classes

- Today you write **one small class** with an `__init__` and **one method**. That's it
- Classes get tricky fast (inheritance, `self` confusion), and that is **not this course**
- We use them only to *bundle* data with behavior, nothing deeper

...

💡 Tip

If the `self` keyword feels odd right now, that's completely normal. Copy the shape from the worked example. The intuition follows the practice.

Your turn: 5-10 minutes

Open the exercise (scan the QR or type the link):

python.tobiasvlcek.com/notebooks/ex_03_c/



First **predict** what happens, then run it.

To the Lab

After the break: the lab

- Head to the lab notebook: [Episode 3: The Copy-Paste Soup](#)
- You'll write the fee and tip functions, fix a function that forgets to `return`, give a parameter a default, and build the startup's first `Order` class, ending in the **Tip Calculator Championship**
- It runs entirely in your browser: no setup, just click and code

...

! Important

Download your `.py` before you leave. Closing the tab without downloading loses your work, and downloading is exactly how you just handed in the checkpoint.

Wrap-up

Three things to remember

1. A **function** is written once with `def`, takes **parameters**, and hands a value back with `return`: `print` shows, `return` gives back (no `return` → `None`)
2. Names inside a function are **local**: it can't quietly change your variables, which is exactly what makes it safe to reuse
3. A **class** bundles data (on `self`, via `__init__`) with behavior (methods). You'll write just one small one today

...

i Note

Next time, Episode 4: the menu outgrows Tobi's seventeen loose variables (`price1`, `price2`, `price_final_FINAL2`), and nobody can find anything. We give the data a shape: lists and dictionaries.

Literature

Books to start with

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. [Link to free online version](#)
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.

...

i Note

Nothing new here, but these are still great books to start with!

...

For more, see the [literature list](#) of this course.